

EXAMPLES

GALE - SHAPLEY ALGORITHM

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CODE

I. Code

II. Example 1

III. Example 2

IV. Example 3

V. Example 4

VI. Conclusion

```
int main() {

    vector<vector<int>> menPref(n, vector<int>(n));
    vector<vector<int>> womenPref(n, vector<int>(n));

    queue<int> freeMen;
    for(int i = 0; i < n; i++) freeMen.push(i);

    vector<int> menPartner(n, -1);
    vector<int> womenPartner(n, -1);

    vector<int> nextProposal(n, 0);

    vector<vector<int>> menRank(n, vector<int>(n));
    for (int i = 0; i < n; i++) {
        int rank=n-1;
        for (int j = 0; j < n; j++) {
            menRank[i][womenPref[i][j]]=rank--;
        }
    }
}
```

```
while(!freeMen.empty()){
    int currm=freeMen.front();
    freeMen.pop();

    int currw = menPref[currm][nextProposal[currm]];

    if(womenPartner[currw]==-1){
        menPartner[currm]=currw;           Unmatched woman
        womenPartner[currw]=currm;
    }

    else
        if(menRank[currw][currm] < menRank[currw][womenPartner[currw]]){
            freeMen.push(currm);           Matched woman and no change
        }
        else{
            menPartner[womenPartner[currw]]=-1;
            freeMen.push(womenPartner[currw]); //unmatch Matched woman
                                                    and wants to
                                                    change
            menPartner[currm]=currw;
            womenPartner[currw]=currm;      //match
        }
    }

    nextProposal[currm]++;
}
```

CODE

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Programiz C++ Online Compiler

main.cpp

Share

Run

```
1  #include <iostream>
2  using namespace std;
3  #include <vector>
4  #include <string>
5  #include <queue>
6
7
8
9
10 int main() {
11
12     int n;
13     cin >> n;
14
15     vector<vector<int>> menPref(n, vector<int>(n));
16     vector<vector<int>> womenPref(n, vector<int>(n));
17
18     // Input first matrix
19     for (int i = 0; i < n; i++) {
20         for (int j = 0; j < n; j++) {
21             cin >> menPref[i][j];    // men i women j
22         }
23     }
24
25     // Input second matrix
26     for (int i = 0; i < n; i++) {
```

Output

4
1 0 2 3
2 3 1 0
2 1 3 0
0 3 2 1

2 3 1 0
2 1 0 3
0 1 2 3
3 2 1 0
m0 --- w2
m1 --- w3
m2 --- w1
m3 --- w0

=== Code Execution Successful ===

Output

3
0 1 2
0 2 1
1 0 2

2 0 1
1 2 0
0 2 1
m0 --- w0
m1 --- w2
m2 --- w1

=== Code Execution Successful ===

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EXAMPLE 1

A	X	W	Y	Z
B	Y	Z	X	W
C	Y	X	Z	W
D	W	Z	Y	X

W	C	D	B	A
X	C	B	A	D
Y	A	B	C	D
Z	D	C	B	A

Stable Matching :

A-----Y
B-----Z
C-----X
D-----W

Swap Y and W in A

A	X	Y	W	Z
B	Y	Z	X	W
C	Y	X	Z	W
D	W	Z	Y	X

W	C	D	B	A
X	C	B	A	D
Y	A	B	C	D
Z	D	C	B	A

No blocking pairs

EXAMPLE 2

- I. Code
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- IV. Example 3
- V. Example 4
- VI. Conclusion

A	X	W	Y	Z
B	Y	Z	X	W
C	Y	X	Z	W
D	W	Z	Y	X



W	C	D	B	A
X	C	B	A	D
Y	A	B	C	D
Z	D	C	B	A

Stable Matching :

A-----Y
B-----Z
C-----X
D-----W

Swap Z and W in D

A	X	W	Y	Z
B	Y	Z	X	W
C	Y	X	Z	W
D	Z	W	Y	X

W	C	D	B	A
X	C	B	A	D
Y	A	B	C	D
Z	D	C	B	A

Blocking pair D---Z

A	X	W	Y	Z
B	Y	Z	X	W
C	Y	X	Z	W
D	Z	W	Y	X

W	C	D	B	A
X	C	B	A	D
Y	A	B	C	D
Z	D	C	B	A

Stable Matching :

A-----Y
B-----W
C-----X
D-----Z

1 Update

EXAMPLE 3

A	W	X	Y	Z
B	X	Y	Z	W
C	Y	X	Z	W
D	Z	Y	X	W

W	C	A	B	D
X	D	A	B	C
Y	A	B	C	D
Z	C	B	A	D

Stable Matching :

A-----Y
B-----Z
C-----X
D-----W

Swap Z and W in A

A	Z	X	Y	W
B	X	Y	Z	W
C	Y	X	Z	W
D	Z	Y	X	W

W	C	A	B	D
X	D	A	B	C
Y	A	B	C	D
Z	C	B	A	D

Blocking pair A---Z

A	Z	X	Y	W
B	X	Y	Z	W
C	Y	X	Z	W
D	Z	Y	X	W

W	C	A	B	D
X	D	A	B	C
Y	A	B	C	D
Z	C	B	A	D

Blocking Pair D---X

EXAMPLE 3

A	Z	X	Y	W
B	X	Y	Z	W
C	Y	X	Z	W
D	Z	Y	X	W

W	C	A	B	D
X	D	A	B	C
Y	A	B	C	D
Z	C	B	A	D

Blocking pair B---Y

A	Z	X	Y	W
B	X	Y	Z	W
C	Y	X	Z	W
D	Z	Y	X	W

W	C	A	B	D
X	D	A	B	C
Y	A	B	C	D
Z	C	B	A	D

Blocking pair C---Z

A	Z	X	Y	W
B	X	Y	Z	W
C	Y	X	Z	W
D	Z	Y	X	W

W	C	A	B	D
X	D	A	B	C
Y	A	B	C	D
Z	C	B	A	D

Blocking Pair A---Y

EXAMPLE 3

A	Z	X	Y	W
B	X	Y	Z	W
C	Y	X	Z	W
D	Z	Y	X	W

W	C	A	B	D
X	D	A	B	C
Y	A	B	C	D
Z	C	B	A	D

Stable Matching :

A-----Y

B-----W

C-----Z

D-----X

5 updates

W	C	A	B	D
X	D	A	B	C
Y	A	B	C	D
Z	C	B	A	D

EXAMPLE 4



A	W	X	Y	Z
B	Z	X	W	Y
C	Y	Z	W	X
D	X	Y	Z	W

W	B	C	D	A
X	A	B	C	D
Y	D	A	B	C
Z	C	D	A	B

Swap Z and W in A

A	Z	X	Y	W
B	Z	X	W	Y
C	Y	Z	W	X
D	X	Y	Z	W

W	B	C	D	A
X	A	B	C	D
Y	D	A	B	C
Z	C	D	A	B

Blocking pair A---Z

A	Z	X	Y	W
B	Z	X	W	Y
C	Y	Z	W	X
D	X	Y	Z	W

W	B	C	D	A
X	A	B	C	D
Y	D	A	B	C
Z	C	D	A	B

Blocking pair B---X

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IV.	Example 3
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EXAMPLE 4

A	Z	X	Y	W
B	Z	X	W	Y
C	Y	Z	W	X
D	X	Y	Z	W

W	B	C	D	A
X	A	B	C	D
Y	D	A	B	C
Z	C	D	A	B

Blocking Pair D---Y

A	Z	X	Y	W
B	Z	X	W	Y
C	Y	Z	W	X
D	X	Y	Z	W

W	B	C	D	A
X	A	B	C	D
Y	D	A	B	C
Z	C	D	A	B

Blocking pair C---Z

A	Z	X	Y	W
B	Z	X	W	Y
C	Y	Z	W	X
D	X	Y	Z	W

W	B	C	D	A
X	A	B	C	D
Y	D	A	B	C
Z	C	D	A	B

Blocking pair A---X

EXAMPLE 4

A	Z	X	Y	W
B	Z	X	W	Y
C	Y	Z	W	X
D	X	Y	Z	W

W	B	C	D	A
X	A	B	C	D
Y	D	A	B	C
Z	C	D	A	B

Stable Matching :

A-----X

B-----W

C-----Z

D-----Y

5 updates

W	B	C	D	A
X	A	B	C	D
Y	D	A	B	C
Z	C	D	A	B

CONCLUSION

- With one exchange in men's preference list a loop of blocking pairs couldn't exist.
- In one case it is found that the modified matching turned out to be women optimum stable matching.
- It is possible to modify the matching in $O(n^2)$ time .

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THANK YOU

ANY QUESTIONS